

Notes: Bolton, Freixas, and Shapiro (JF, 2012)

The Credit Ratings Game

Contents

1 Big picture

2 Model environment and key objects

- 2.1 Agents, investment, and uncertainty
- 2.2 CRA signal technology
- 2.3 Investor demand and trusting versus sophisticated investors
- 2.4 Issuer shopping and uniform pricing
- 2.5 Reputation costs
- 2.6 Assumptions that create a meaningful role for CRAs
- 2.7 Valuation objects
- 2.8 Timing of the game

3 Models and theoretical specifications (all equations + interpretation)

- 3.1 Economic mechanism and hypotheses
- 3.2 Monopoly CRA: reporting rule
- 3.3 Monopoly CRA: equilibrium fee and information regime
- 3.4 Duopoly CRA: additional notation and assumptions
- 3.5 Duopoly CRA: reporting rule and equilibrium
- 3.6 Efficiency benchmarks
- 3.7 Monopoly versus duopoly efficiency
- 3.8 Proposition 3: truth-telling monopoly dominates truth-telling duopoly
- 3.9 When can competition help?
- 3.10 Endogenous reputation: two-period extension
- 3.11 Regulation and policy design

4	Identification and comparison logic (what makes the results credible)	
4.1	What identifies the main result in a theory paper	
4.2	Why monopoly can dominate duopoly	
4.3	Why trusting investors are essential	
4.4	Why the result is not just about CRA dishonesty	
4.5	Why signal precision has an ambiguous effect	
4.6	Why the two-period reputation extension is informative	
4.7	What the paper can and cannot distinguish	
4.8	Relation to adjacent literatures	
5	Main propositions and empirical implications	
5.1	Proposition 1: monopoly CRA may inflate ratings	
5.2	Proposition 2: duopoly does not solve the problem	
5.3	Proposition 3: truth-telling monopoly strictly dominates truth-telling duopoly	
5.4	Lemma 3: competition helps only if it changes the information regime	
5.5	Proposition 4: endogenous reputation makes the monopoly case stronger	
5.6	Regulatory implications	
5.7	Empirical implications and the paper’s evidence review	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** How do conflicts of interest in the credit rating industry affect ratings quality, market efficiency, and the desirability of competition among credit rating agencies (CRAs)?
- **Setting.** The paper builds a theoretical model motivated by the structured-finance boom and bust. Issuers pay CRAs, issuers can suppress unfavorable ratings, some investors understand these incentives, and others naively take ratings at face value.
- **Core trade-off.** Competition can in principle produce more information because two ratings may be better than one. But competition also gives issuers more opportunities to shop for favorable ratings and target trusting investors.
- **Main conceptual contribution.** The paper combines three frictions in one framework:
 - CRA incentives to understate risk to attract business,
 - issuer ratings shopping,
 - heterogeneous investors, split between sophisticated and trusting clienteles.
- **Main result under monopoly.** Even a monopolist CRA may inflate ratings if the current gain from pleasing the issuer exceeds the expected reputation cost of being caught.
- **Main result under duopoly.** A duopoly may produce more raw information, but it generally reduces efficiency because it creates extra opportunities for issuer shopping. When both market structures are in the same informational regime, monopoly strictly dominates duopoly.
- **Key mechanism.** A second rating is not always socially useful. When an issuer can selectively disclose only the favorable rating, the extra report often helps mislead trusting investors rather than inform sophisticated ones.
- **Boom-time implication.** Ratings inflation is more likely in booms, when trusting investors are more prevalent, fees are larger, and ex post detection is less likely because defaults are less frequent.
- **Dynamic result.** When reputation is endogenized in a simple two-period extension, monopoly and duopoly end up in the same information regime. This makes monopoly always more efficient than duopoly in that extension.
- **Regulatory message.** Up-front CRA fees can eliminate one conflict of interest, but do not eliminate ratings shopping. Automatic disclosure of solicited ratings is needed to address shopping, and oversight of methodology is needed to prevent underinvestment in rating precision.
- **Economic takeaway.** More competition is not always better in certification markets. When issuers can shop and some investors are trusting, entry can lower welfare rather than raise it.

2 Model environment and key objects

2.1 Agents, investment, and uncertainty

The model has three types of risk-neutral agents: issuers, CRAs, and investors. The issuer seeks funds for an investment. The investment can be either good or bad.

- A **good** investment never defaults.
- A **bad** investment defaults with probability $p > 0$.
- If the investment does not default, each unit pays return R .

Ex ante, all agents believe the investment is good with probability $1/2$.

Interpretation. The paper creates a role for CRAs by assuming that investors start with only coarse beliefs about project quality. The CRA can potentially raise efficiency by separating good from bad issues.

2.2 CRA signal technology

A CRA receives a private signal $\theta \in \{g, b\}$ about the investment's type $\omega \in \{g, b\}$ with precision

$$\Pr(\theta = g \mid \omega = g) = \Pr(\theta = b \mid \omega = b) = e, \quad (1)$$

where $e \in (1/2, 1)$.

Interpretation. e measures the informativeness of ratings technology. At $e = 1/2$, the signal is useless. As e rises, the CRA's rating becomes more informative, but that also changes the gains from misreporting and the probability of being caught.

2.3 Investor demand and trusting versus sophisticated investors

Investors can buy either one unit or two units of the investment. Their reservation return increases with investment size:

- the first unit requires return u ,
- the second unit requires return U , where $U > u$.

There are two investor types:

- a fraction $1 - \alpha$ are **sophisticated**; they understand issuer and CRA incentives and update rationally;
- a fraction α are **trusting**; they take ratings at face value and retain prior beliefs when no rating is disclosed.

Interpretation. The entire paper turns on the coexistence of sophisticated and trusting investors. If everyone were sophisticated, issuer shopping would not create the same welfare losses.

2.4 Issuer shopping and uniform pricing

After observing the CRA's report, the issuer decides whether to buy and disclose it. If the issuer refuses to buy the report, no rating is published. The issuer then sets a uniform price T per unit of the investment.

Interpretation. Selective disclosure is the shopping margin. It is not that CRAs must openly lie every time. Even with truthful CRAs, issuers can still distort what the market sees by publishing only the favorable reports.

2.5 Reputation costs

If a CRA lies by reporting $m = G$ after receiving a bad signal and the investment later defaults, investors infer that the CRA misled them and punish it in future periods. The discounted value of lost future business is summarized by a reputation cost ρ in monopoly and ρ^D in duopoly.

The paper first treats these reputation costs as exogenous. Later it endogenizes them in a two-period model.

Interpretation. The CRA's incentive problem is simple: short-run fee income pushes toward inflated ratings, while reputation loss pushes toward truthful ratings.

2.6 Assumptions that create a meaningful role for CRAs

The paper assumes:

$$(1 - p)R > u, \quad (2)$$

$$(1 - (1 - e)p)R > U, \quad (3)$$

$$\left(1 - \frac{p}{2}\right) R < U. \quad (4)$$

Interpretation.

- The first condition says that if an investor knows the investment is bad, buying one unit is still worthwhile.
- The second says that if an investor gets a reliable good signal, buying two units is worthwhile.
- The third is the key information friction: with only ex ante beliefs, investors will *not* buy two units. So certification can create real value by expanding funding for good projects.

2.7 Valuation objects

To simplify the analysis, the paper defines

$$V^G = (1 - (1 - e)p)R - U, \quad (5)$$

$$V^B = (1 - ep)R - u, \quad (6)$$

$$V^0 = \left(1 - \frac{p}{2}\right) R - u. \quad (7)$$

Here:

- V^G is the marginal value of a positive report,
- V^B is the marginal value of a negative report,
- V^0 is the marginal value under prior beliefs.

With two CRAs, the analogous objects are

$$V^{GG} = \left(1 - \frac{(1 - e)^2}{(1 - e)^2 + e^2} p\right) R - U, \quad (8)$$

$$V^{BB} = \left(1 - \frac{e^2}{(1-e)^2 + e^2 p}\right) R - u. \quad (9)$$

Interpretation. These objects compress the informational environment into a few payoff terms. Once they are defined, the CRA's reporting and fee-setting problem becomes easy to state.

2.8 Timing of the game

Monopoly:

1. The CRA posts a fee ϕ .
2. The CRA receives a signal and reports $m \in \{G, B\}$.
3. The issuer decides whether to purchase and disclose the report, then sets price T .
4. Investors observe T and the disclosed rating, if any, and choose how much to buy.
5. The investment return is realized.

Duopoly:

1. Each CRA $k \in \{1, 2\}$ posts fee ϕ_k .
2. Each CRA receives its signal and reports $m \in \{G, B\}$.
3. The issuer chooses whether to purchase one, both, or neither report, then sets price T .
4. Investors observe the purchased report set and decide how much to buy.
5. The investment return is realized.

Interpretation. The timing formalizes why shopping matters: the issuer sees the rating before deciding whether the market gets to see it.

3 Models and theoretical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper studies two frictions together.

- **Conflict of interest:** because issuers pay for ratings, CRAs may want to overstate quality to attract business.
- **Ratings shopping:** because issuers may publish only favorable ratings, competition can make the disclosure problem worse.

The central theoretical question is whether the extra information created by competition outweighs the extra distortion created by shopping.

Interpretation. The answer is not obvious ex ante. Two CRAs can create more information, but that information may never reach the market in a symmetric way.

3.2 Monopoly CRA: reporting rule

Given fee ϕ , the monopoly CRA compares the fee from publishing a favorable rating with the expected reputation loss from being caught lying.

Lemma 1.

$$\phi > e\rho \implies \text{the CRA inflates and always reports } G, \quad (10)$$

while

$$0 < \phi < e\rho \implies \text{the CRA tells the truth.} \quad (11)$$

Interpretation. The monopoly CRA has a simple threshold rule. If current fee income is larger than the expected punishment, it always inflates. Otherwise it reports truthfully.

3.3 Monopoly CRA: equilibrium fee and information regime

Proposition 1. The monopoly equilibrium is:

1. If

$$\alpha 2V^G - V^0 > e\rho, \quad (12)$$

the CRA inflates, sets

$$\phi = \alpha 2V^G - V^0, \quad (13)$$

and earns profits

$$\alpha 2V^G - V^0 + \left(1 - \frac{e\rho}{2}\right) \rho. \quad (14)$$

2. If

$$\alpha 2V^G - V^0 < e\rho, \quad (15)$$

the CRA tells the truth, sets

$$\phi = \min\{2V^G - \max(\alpha V^0, V^B), e\rho\}, \quad (16)$$

and earns profits

$$\frac{1}{2} \min\{2V^G - \max(\alpha V^0, V^B), e\rho\} + \rho. \quad (17)$$

Interpretation. The relevant cutoff is

$$\alpha 2V^G - V^0 - e\rho. \quad (18)$$

Ratings inflation becomes more likely when the trusting audience α is larger and reputation cost ρ is smaller. Precision e has an ambiguous effect because it both raises the current payoff from a favorable rating and raises the probability of being caught when lying.

3.4 Duopoly CRA: additional notation and assumptions

With two CRAs, the issuer may want to buy both ratings because two signals can be more informative than one. The paper imposes three assumptions to simplify the equilibrium analysis.

$$\text{Assumption 5: } \alpha 2V^G - V^0 > 2(V^{GG} - V^G), \quad (19)$$

$$\text{Assumption 6: } \alpha 2V^G - V^0 - \min\{2(V^{GG} - V^G), ep\rho^D\} < e^2 p\rho^D, \quad (20)$$

$$\text{Assumption 7: } \frac{V^0}{2V^G} < \frac{V^{BB}}{V^0}. \quad (21)$$

Interpretation.

- Assumption 5 says the first favorable report matters more than the second. This builds decreasing returns to good news into the duopoly case.
- Assumption 6 guarantees existence of the truth-telling duopoly equilibrium by ruling out a profitable unilateral deviation to inflated ratings.
- Assumption 7 simplifies the pricing region in which issuers target trusting investors after bad news.

3.5 Duopoly CRA: reporting rule and equilibrium

Lemma 2. For a given fee vector,

$$\phi_k > ep\rho^D \implies \text{CRA } k \text{ inflates and always reports } G, \quad (22)$$

and

$$\phi_k < ep\rho^D \implies \text{CRA } k \text{ tells the truth.} \quad (23)$$

Proposition 2. The duopoly equilibrium is:

1. If

$$\alpha 2(V^{GG} - V^G) > ep\rho^D, \quad (24)$$

both CRAs inflate, each sets

$$\phi_k = \alpha 2(V^{GG} - V^G), \quad (25)$$

and profits are

$$\alpha 2(V^{GG} - V^G) + \left(1 - \frac{ep}{2}\right) \rho^D. \quad (26)$$

2. If

$$\alpha 2(V^{GG} - V^G) < ep\rho^D, \quad (27)$$

both CRAs tell the truth, each sets

$$\phi_k = \min\{2(V^{GG} - V^G), ep\rho^D\}, \quad (28)$$

and profits are

$$\frac{1}{2} \min\{2(V^{GG} - V^G), ep\rho^D\} + \rho^D. \quad (29)$$

Interpretation. The duopoly threshold looks like the monopoly threshold, but the current gain from manipulation is based on the extra value of going from one favorable rating to two favorable ratings. Because the second good rating has lower marginal value, the direct gain from inflation can be smaller under duopoly. But this does not mean duopoly is more efficient; shopping may still dominate.

3.6 Efficiency benchmarks

The paper evaluates ex ante total surplus from the perspective of a sophisticated observer. It first defines two benchmarks.

First-best surplus:

$$W_{FB} = \frac{1}{2}(2R - u - U) + \frac{1}{2}((1 - p)R - u) = V^0 + \frac{1}{2}(R - U). \quad (30)$$

No-CRA market solution:

$$W_0 = V^0. \quad (31)$$

Interpretation. The maximum social gain from certification is the extra second unit of investment that can be sold when the issue is genuinely good.

3.7 Monopoly versus duopoly efficiency

The paper derives surplus in all regimes. Two expressions are especially important.

Monopoly with ratings inflation:

$$W_M^G = \alpha[V^0 + (V^0 + u - U)]. \quad (32)$$

Monopoly with truthful ratings:

$$W_{MT1} = V^0 + \frac{1}{2}V^G, \quad (33)$$

with a lower-surplus variant

$$W_{MT2} = V^0 + \frac{1}{2}V^G - \frac{1 - \alpha}{2}V^B \quad (34)$$

when issuers target only trusting investors after no report.

Duopoly with truthful ratings:

$$W_{DT1} = (e^2 + 2e(1 - e)\alpha + (1 - e)^2)V^0 + \frac{1}{2}(e^2 - (1 - e)^2)(R - U) \quad (35)$$

$$+ (2e(1 - e)\alpha + (1 - e)^2)(V^0 + u - U), \quad (36)$$

with a lower-surplus variant

$$W_{DT2} = W_{DT1} - \frac{1 - \alpha}{2}[(1 - e)^2(R - u) + e^2((1 - p)R - u)]. \quad (37)$$

Interpretation. The formulas are messy, but the economics is clean: duopoly creates more states in which issuers can cherry-pick reports and sell only to trusting investors at distorted prices.

3.8 Proposition 3: truth-telling monopoly dominates truth-telling duopoly

Proposition 3.

A truth-telling monopoly strictly dominates a truth-telling duopoly. (38)

Interpretation. This is the paper's most striking welfare result. Even when both CRAs tell the truth, duopoly is worse because the issuer gets more ways to exploit trusting investors. In monopoly, shopping happens only when the single CRA reports bad news. In duopoly, shopping can occur when one CRA reports good and the other bad, or when both report bad. The second report's extra information is frequently wasted.

3.9 When can competition help?

The paper also proves a useful intermediate result.

Lemma 3. A truth-telling duopoly yields higher total surplus than a monopoly that inflates ratings.

Interpretation. Competition can help only if it changes the information regime from inflated ratings to truthful ratings. But conditional on being in the same regime, monopoly is better.

3.10 Endogenous reputation: two-period extension

To endogenize reputation, the paper introduces a two-period model with discount-like weight β on second-period profits. In period 2 there are no future reputation concerns, so the CRA always inflates. This makes future profits the punishment for first-period lying.

For monopoly,

$$\rho = \beta(\alpha 2V^G - V^0), \quad (39)$$

so first-period inflation occurs iff

$$\beta < \frac{1}{ep}. \quad (40)$$

For duopoly,

$$\rho^D = \beta\alpha 2(V^{GG} - V^G), \quad (41)$$

which yields the same inflation condition:

$$\beta < \frac{1}{ep}. \quad (42)$$

Proposition 4.

$$\beta \geq \frac{1}{ep} \iff \text{truth-telling in both monopoly and duopoly}, \quad (43)$$

$$\beta < \frac{1}{ep} \iff \text{ratings inflation in both monopoly and duopoly}. \quad (44)$$

Interpretation. Once reputation is endogenous in this simple way, monopoly and duopoly always end up in the same information regime. Since monopoly is already more efficient conditional on the regime, endogenizing reputation strengthens the case for monopoly.

3.11 Regulation and policy design

The model is used to interpret several regulatory ideas.

- **Up-front fees (the Cuomo plan):** can eliminate the issuer's ability to reward a favorable rating, so it removes one source of ratings inflation.
- **Automatic disclosure of all solicited ratings:** is needed to eliminate shopping.
- **Methodology oversight:** is needed because up-front payment may induce CRA underinvestment in analytical precision if precision is chosen after payment and is noncontractible.
- **Track-record transparency and liability:** increase effective reputation costs and make truth telling more likely.

Interpretation. The paper's regulatory message is not merely "ban conflicts of interest." It is a three-part package: change compensation, force disclosure, and monitor analytical quality.

4 Identification and comparison logic (what makes the results credible)

4.1 What identifies the main result in a theory paper

This is not an empirical identification exercise. The paper identifies the welfare ranking by solving the same underlying environment under different market structures and comparing equilibrium outcomes.

Interpretation. The analogue of identification here is a clean mechanism comparison: what changes when we move from one CRA to two CRAs, holding fixed the signal technology, investor composition, and issuer behavior?

4.2 Why monopoly can dominate duopoly

The most important logical step is that competition does two things at once:

- it may produce more information,
- but it also increases the issuer's opportunity set for selective disclosure.

Interpretation. The paper's contribution is to show that the second effect can dominate the first. More information is not automatically socially useful if the issuer controls what reaches investors.

4.3 Why trusting investors are essential

If all investors were sophisticated, shopping would not generate the same distortion because the market would understand the selection effect. The welfare losses from duopoly therefore hinge on the presence of trusting investors.

Interpretation. The paper is fundamentally a model of certification in a market with heterogeneous sophistication. Without naive investors, monopoly-versus-duopoly would look much more standard.

4.4 Why the result is not just about CRA dishonesty

A key insight is that inefficiency can arise even if CRAs tell the truth. The duopoly result is not driven solely by direct lying. It also comes from the issuer's ability to disclose only the ratings it likes.

Interpretation. This matters because it shifts attention away from "bad rating agencies" alone and toward market design. Even truthful intermediaries can generate bad outcomes if disclosure is selective.

4.5 Why signal precision has an ambiguous effect

Higher signal precision does two things:

- it raises the value of a favorable report to trusting investors, which increases current manipulation rents,
- but it also raises the probability that a lie is detected after default, which increases expected punishment.

Interpretation. The paper therefore predicts that better models do not mechanically fix ratings inflation. Precision changes both temptation and discipline.

4.6 Why the two-period reputation extension is informative

The dynamic extension is simple, but it isolates an important point: once future business constitutes the reputation cost, the same threshold governs truthful behavior in monopoly and duopoly.

Interpretation. This means the basic welfare comparison is not an artifact of exogenous reputation parameters. If anything, endogenizing reputation makes the monopoly advantage sharper.

4.7 What the paper can and cannot distinguish

The paper is strongest on comparative logic and weaker on institutional detail.

- It can distinguish the direct conflict-of-interest problem from the shopping problem.
- It can show why competition may worsen shopping.
- It cannot fully model all industry details, such as many-period monitoring, fine rating gradations, or the exact legal process through which liability bites.

Interpretation. The model is deliberately stripped down to two rating levels and short-horizon welfare. That simplicity is what gives the clean monopoly-versus-duopoly result, but also limits how far one should push the quantitative interpretation.

4.8 Relation to adjacent literatures

The paper sits at the intersection of several literatures:

- information-intermediary models with endogenous reputation,
- strategic disclosure and certification,
- industrial-organization models of competition under asymmetric information,
- and the post-crisis literature on structured-finance ratings and shopping.

Interpretation. The paper's distinct contribution is to combine issuer-side shopping, CRA-side conflicts, and investor naivete in one tractable framework.

5 Main propositions and empirical implications

5.1 Proposition 1: monopoly CRA may inflate ratings

Read:

- The monopoly CRA inflates whenever the current fee gain $\alpha 2V^G - V^0$ exceeds expected reputation loss epp .
- Otherwise it tells the truth.

Economic message. Even a monopolist with reputation concerns may systematically overrate issues. Monopoly does not solve the issuer-pays conflict by itself.

5.2 Proposition 2: duopoly does not solve the problem

Read:

- In duopoly, both CRAs are in the same regime: either both inflate or both tell the truth.
- The duopoly cutoff compares $\alpha 2(V^{GG} - V^G)$ with epp^D .

Economic message. Competition changes the fee game but not in a way that guarantees efficient disclosure. The issuer's ability to shop remains central.

5.3 Proposition 3: truth-telling monopoly strictly dominates truth-telling duopoly

Read:

- This is the paper's headline welfare result.
- Conditional on both structures being truthful, monopoly has higher surplus than duopoly.

Economic message. Extra ratings are not necessarily socially beneficial. In this model, the second rating frequently becomes a tool of issuer selection rather than a genuine information gain for the market.

5.4 Lemma 3: competition helps only if it changes the information regime

Read:

- A truth-telling duopoly can beat a monopoly that inflates ratings.
- But this only happens when duopoly shifts the market into a more truthful regime.

Economic message. The benefit of competition is regime-switching, not competition per se. If monopoly and duopoly are both truthful, monopoly wins. If both inflate, monopoly weakly wins again.

5.5 Proposition 4: endogenous reputation makes the monopoly case stronger

Read:

- In the two-period extension, the same cutoff $\beta \geq 1/ep$ determines truth-telling in both market structures.
- Therefore monopoly always dominates duopoly in that extension.

Economic message. Endogenous reputation does not rescue competition. It instead aligns the informational regime across market structures and leaves monopoly with the efficiency edge.

5.6 Regulatory implications

Read:

- Up-front fees eliminate one channel of ratings inflation.
- Automatic disclosure of solicited ratings is needed to eliminate shopping.
- Oversight of rating methodology is needed because up-front payment may otherwise reduce effort in producing accurate signals.
- More transparency and legal liability can raise effective reputation costs.

Economic message. The optimal reform is not a single lever. It is a package that attacks all three problems: direct conflicts, selective disclosure, and weak incentives to maintain model quality.

5.7 Empirical implications and the paper's evidence review

The paper closes by mapping theory to evidence. The main empirical implications are:

- ratings quality should be worse in booms;
- shopping should be more important when ratings models are less precise;
- yields should depend more on ratings when investors are more trusting;
- fewer published ratings should predict worse ex post performance.

The authors discuss several pieces of supporting evidence:

- **Becker and Milbourn (2011):** greater competition from Fitch coincided with worse corporate bond ratings quality.
- **Ashcraft, Goldsmith-Pinkham, and Vickery:** structured-finance ratings were least accurate near the peak of the real-estate boom.
- **He, Qian, and Strahan:** larger structured-product issuers received more favorable ratings.
- **Adelino:** AAA-tranche investors appear especially reliant on ratings, consistent with trusting behavior.
- **Kisgen and Strahan; Bongaerts, Cremers, and Goetzmann:** ratings matter especially where regulation or tie-breaking creates mechanical demand for them.

Economic message. The paper is theoretical, but its empirical section makes clear that the model is meant to explain concrete post-crisis facts: shopping, boom-time inflation, naive demand, and the ambiguous role of competition.

6 Short summary

- The paper models CRA behavior in a setting with issuer-paid ratings, ratings shopping, reputation concerns, and heterogeneous investors.
- A monopoly CRA inflates ratings when current fee gains exceed expected reputation losses.
- A duopoly may create more information, but it also creates more opportunities for shopping.
- Conditional on being in the same information regime, monopoly is more efficient than duopoly.
- In a two-period extension with endogenous reputation, monopoly and duopoly enter the same regime, so monopoly always dominates.
- Up-front fees alone are not enough. The efficient reform package combines up-front payment, automatic disclosure of solicited ratings, and oversight of rating methodology.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that competition among CRAs can lower rather than raise efficiency because it facilitates issuer shopping and allows issuers to exploit trusting investors. This conclusion survives even after the model is extended to include endogenous reputation in a simple dynamic way.

7.2 Economic intuition

The natural intuition that more certifiers create better information is incomplete. Information only improves market outcomes when it is disclosed symmetrically and interpreted correctly. When issuers can suppress bad news and some investors naively trust ratings, extra certifiers often increase the scope for manipulation more than the amount of useful information.

7.3 Takeaway

This paper is a clean reminder that the industrial organization of information production matters. In markets for certification, the issuer-pays model, selective disclosure, and heterogeneous investor sophistication can combine to make competition socially harmful. The right policy response is therefore not simply “more competition,” but rather a redesign of disclosure rules and incentives.